Software Development Lifecycle (SDLC) Overview

Applicable Roles: DEVELOPER, QA, PM, DEVOPS, PO

1. Planning

The planning phase involves requirement gathering, feasibility analysis, and project planning. Key activities:

- Stakeholder meetings to define scope
- Resource allocation and timeline estimation
- Risk assessment and mitigation planning

2. Requirements Analysis

Gather and document detailed functional and non-functional requirements:

- Create BRD (Business Requirements Document)
- Define user stories and acceptance criteria
- Get sign-off from stakeholders

3. Design

System architecture and design specifications:

- Create HLD (High-Level Design) and LLD (Low-Level Design)
- Define data models and ERDs
- Design API contracts and interfaces
- Security architecture review

4. Development

Coding and implementation following best practices:

- Follow coding standards and style guides
- Write unit tests (minimum 80% coverage)
- Regular code reviews
- Use feature branches and PR workflow

5. Testing

QA and validation:

- Unit testing (developer)
- Integration testing
- System testing
- UAT (User Acceptance Testing)
- Performance/Load testing

6. Deployment

Release to production:

- Create release notes

- Obtain deployment approval
- Execute deployment plan
- Monitor for issues post-deployment
- Rollback plan in place

7. Maintenance

Ongoing support and improvement:

- Bug fixes and patches
- Performance monitoring
- Regular security updates
- Feature enhancements based on feedback